

- Hide menu
- Week 4 Intro
- Lesson 1: Confidence Intervals
- Lesson 2: Point Estimation, Part I
- Lesson 3: Point Estimation, Part II
- Video: Variance Estimation: Estimator's Unbiased, Part 212 min
- Video: Two Principles for Point Estimation5 min
- Graded Assignment: Learning Check - Point Estimation, ContinuedSubmitted
- Weekly Assignment

Your grade: 100%

Next item →

Learning Check - Point Estimation, Continued

Instructions

Review Learning Objectives

1. Let $X_1, X_2, \dots, X_n$ be i.i.d. Gaussian random variables with mean μ and variance σ^2 . Then, the sample mean $(\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i)$ as a point estimate for the population mean would be...

1 / 1 point

biased

unbiased

Cannot be determined

Help me practice

Let's chat

Correct

$E[\bar{X}] = \mu$, so $E[\bar{X}]$ is unbiased.

Due

Attempts

Try again

Oct 1, 11:59 PM IST

4 left (5 attempts every 24 hours)

Sep 26, 12:44 AM IST

2. Let $X_1, X_2, \dots, X_n \sim \text{Ber}(p)$ i.i.d. random variables with parameter p . Alice and Bob are trying to come up with different estimators for the parameter p . Bob chooses the point estimate to be $\bar{X}_n$ and Alice chooses the estimator $\bar{X}_n$ as the parameter p . Both of them claim that their estimator is unbiased. Which of the following is true?

1 / 1 point

Both Alice's and Bob's estimator are unbiased.

Both Alice's and Bob's estimator are biased.

Alice's estimator is unbiased and Bob's estimator is biased.

Alice's estimator is biased and Bob's estimator is unbiased.

Correct

Since $E[\bar{X}] = \mu$, both the estimators are unbiased. But, $\bar{X}_n$ is not a good estimator compared to the sample mean because $\bar{X}_n$ can only be 0 or 1 in a Bernoulli distribution.

3. When choosing between two estimators for a parameter θ we generally prefer ones that are...

1 / 1 point

Unbiased and high variance

Unbiased and low variance

Biased and high variance

Biased and low variance

Correct

The "Two Principles for Point Estimation" are that we prefer unbiased estimators, and among the unbiased estimators we prefer the one with the minimum variance.

https://www.coursera.org/learn/statistics-for-data-science-essentials/assignment/submission/WACN/learning-check-point-estimation-continued/view-feedback

1/1